CSC 466 Project Update 2

Fixed and Adaptive Input Timeouts for a Phone-Controlled Multiplayer Game

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1. Implementation

The Unity host now supports fixed and adaptive timeouts. Adaptive mode starts at 500 ms and, after four usable arrival intervals, uses the smoothed interval plus four times the smoothed absolute deviation. This is then bounded between 80 and 500ms. The host displays the active threshold and logs it alongside input arrivals and timeout transitions.

2. Unity Experiment

I ran the host and controller prefabs in one Unity Editor process, connected over localhost WebSockets. An Editor-only script supplied movement and paused transmission while keeping the connection open. The controller was configured for 30 Hz. For each mode, I repeated two cases three times:

1) A roughly 200 ms pause while movement was still held
2) A roughly 700 ms pause at release before the controller could send neutral input

Results

Measure	**Fixed**	**Adaptive**
Input messages received	206	206
Mean interval outside pauses (193 per mode)	33.33 ms	33.25 ms
Timeouts during held-input pauses	0 of 3	3 of 3
Timeouts after release	3 of 3	3 of 3
Mean release-to-timeout time	519.5 ms	100.0 ms
Mean timeout duration during held input	0 ms	161.1 ms

Timeouts cleared when input resumed	3 of 3	6 of 6

Adaptive hit is 80 ms floor at all six timeout events. The roughly 420 ms improvement after release came with interruptions during all three held-input pauses. The 161.1 ms figure measures from the host’s timeout to the first resumed input, so it includes the wait for that packet.

5. Limitations and Next Steps

The was a same-machine test at one send rate. Pausing sends created gaps at the application level. The next tests should use separate devices, phone controls, additional send-rates, and controlled network delay and jitter.